

Boston House Price Prediction

Machine Learning Elective Project

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Executive Summary

Housing is an essential need for any person. There are many attributes that affect house pricing worldwide, and those many attributes make it hard to assume prices on the top of your mind. The real estate market is an environment where proper valuation is crucial for buyers, sellers, and third-party members to do business and decision-making. Sellers do not adjust prices based on blank logic or personal desire, nor do buyers purchase houses with random budget and logic. Accurate price predictions allow buyers to not overpay and for sellers to set a reasonable competitive price.

Data and trends are analyzed in order to use accurate values/numbers and past house pricing serves as a good reference to predict future housing prices. The purpose of this report is to use housing data that we have to predict that of which are unknown, also figuring out what attributes are more determinant for house pricing.

Business Problem Overview

The current problem is how to predict the house prices in Boston based on the data that was given to us. Again, each attribute can play a role in determining prices and trends/patterns from known data can be referenced in decision-making. By processing and analyzing the given data, a good predictive model can be made through linear regression. On top of that, the model's performance is also examined and improved using cross-validation to give a clear explanation of the problem.

Data overview

We were given a dataset that contains numerical values of 13 different attributes (variables) for each of the 506 houses located in different suburbs or towns in Boston. The data was checked and contained no missing nor duplicate values.

The 13 attributes of the data are the following.

- CRIM: Crime rate per capita by town
- ZN: Residential land proportion for lots over 25,000 sq.ft.
- INDUS: Proportion of non-retail business acres per town
- CHAS: Borders the Charles River, 0=no, 1=yes
- NOX: Nitric Oxide concentration
- RM: Average number of rooms

- AGE: Proportion of occupied units built before 1940
- DIS: Distance to Boston employment centers
- RAD: Radial highway accessibility
- TAX: Property tax rate per \$10,000
- PTRATIO: Pupil-Teacher ratio by town
- LSTAT: % of population with lower status
- MEDV: Median value of occupied homes in \$1000

As each variable has different range of values, a statistics summary was pulled up to get a better idea of each variable values and what it meant. For example, LSTAT would be a percentage ranging from 0 to 100, while RM would be a flat number or rooms with its own reasonable range.

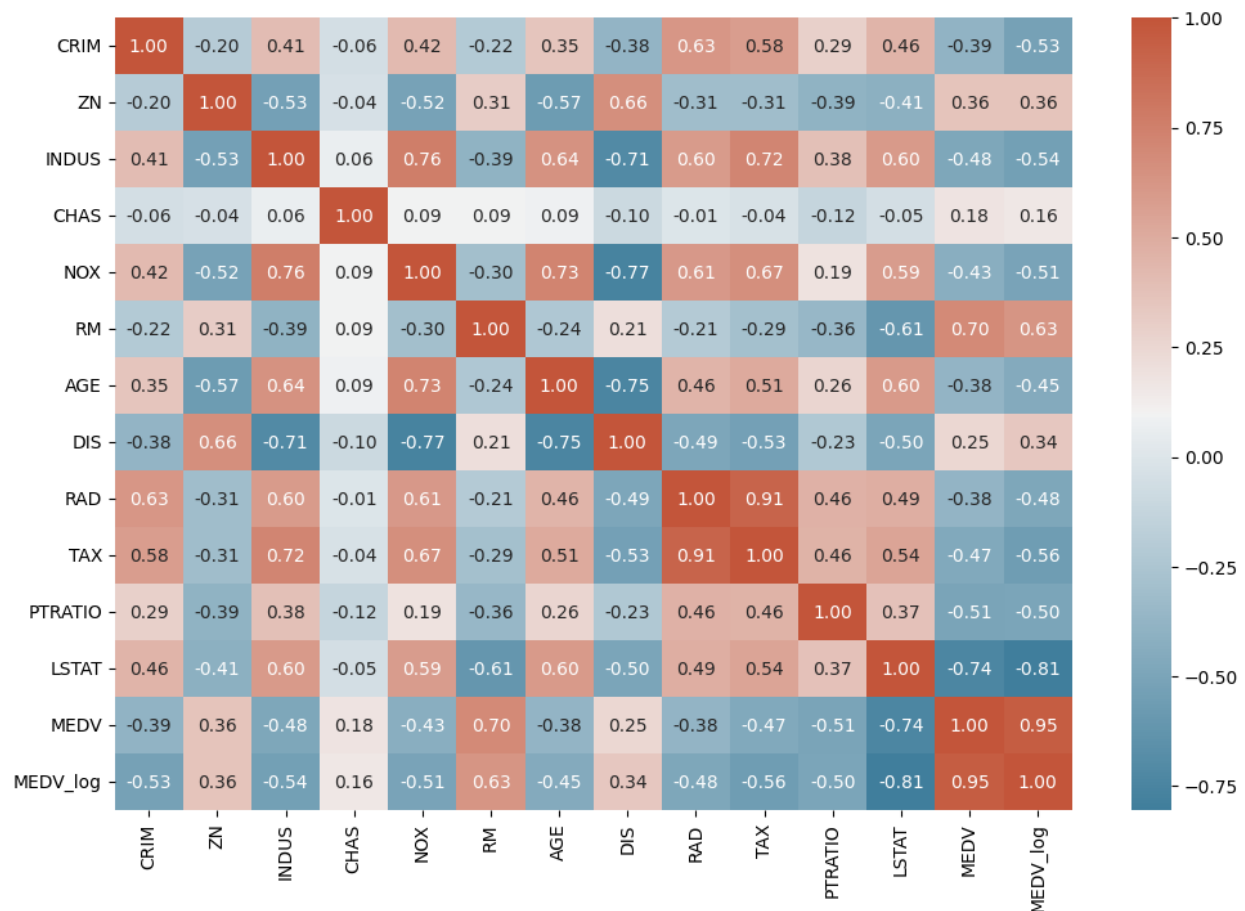
EDA and Data Preprocessing

First off, a univariate analysis was conducted to check the distribution of each variable by creating a histogram with a Kernel Density Estimate (KDE) line for each. Here are the findings.

- CRIM, ZN, CHAS, NOX, DIS, and LSTAT plots showed positively skewed distributions meaning that there were more data points concentrated on the lower end of a value range
- AGE and PTRATIO showed negatively skewed distributions meaning that there were more data points concentrated on the higher end of the value range
- INDUS, RAD, and TAX showed binomial distributions which means there were two peaks in which the data points were concentrated
- MEDV showed a slight positive skew

MEDV, the median value of houses, is the dependent variable in which we are trying to predict. One of the assumptions of the linear regression model is that the residuals are normally distributed. So, a log transformation was applied to better suit this plot for linear regression analysis, creating MEDV_log.

Next, a bivariate analysis was conducted to check for any relationship between the variables. A heatmap was made to give a good illustration of the relationships.



The heatmap gives the correlation values for each variable relationship. First, we checked what variables have strong correlations MEDV. RM has a strong positive correlation value of 0.7 with MEDV, implying that with more rooms, the higher the house price. LSTAT has a strong negative correlation value of -0.74, the house prices are lower in areas with more lower status people. RM and LSTAT are variables that have are likely to have significant effect on house price.

Next, correlations between the independent variables were checked. To better check the relationship between those variables with strong correlation, scatterplots were made to check for any outliers or specific patterns. Here is the summary of the following.

- RAD and TAX have a very strong positive correlation value of 0.91, but the scatterplot showed no trend. This is likely to have occurred due to outliers, and so, after removing the outliers, the correlation value came out to be 0.25.
- TAX and INDUS have a strong positive correlation value of 0.72, and the scatterplot shows a trend of TAX increasing as INDUS increases. This is likely due to property taxes being higher in industrial areas.

- NOX and INDUS have a strong positive correlation value of 0.76, and the scatterplot shows a trend of NOX increasing as INDUS increases. This is likely as there are more people and activity in industrial areas, leading to higher NOX levels.
- NOX and AGE have a strong positive correlation value of 0.73, and the scatterplot shows a trend of NOX increasing as AGE increases. Factors such as traffic and infrastructure can likely cause more NOX in older house areas.
- INDUS and DIS have a strong negative correlation value of -0.71, which shows that the distance to employment centers is shorter in more industrial areas.
- NOX and DIS have a strong negative correlation value of -0.77, and the scatterplot shows a trend of NOX levels decreasing as DIS increases. This is likely due to NOX levels being more produced in industrial zones.
- AGE and DIS have a strong negative correlation value of -0.75, and the scatterplot shows a trend of DIS decreasing as AGE increases. This implies that employment centers are located in older house areas.

Regression Analysis

Correlation between independent variables is not ideal for linear regression models. It can cause multicollinearity, which it makes it hard to see the individual effect of each variable towards the dependent variable. To check for this, the dataset was split using train-test split method and multicollinearity was checked in the training data only. The reason for this is because we are using only the training data to build the predictive model. Checking for multicollinearity on the entire dataset with the test data included spoils the training process leads to overfitting of the model. The Variance Inflation Factor (VIF) was used to check the following.

	feature	VIF		feature	VIF
0	const	535.372593	0	const	532.025529
1	CRIM	1.924114	1	CRIM	1.923159
2	ZN	2.743574	2	ZN	2.483399
3	INDUS	3.999538	3	INDUS	3.270983
4	CHAS	1.076564	4	CHAS	1.050708
5	NOX	4.396157	5	NOX	4.361847
6	RM	1.860950	6	RM	1.857918
7	AGE	3.150170	7	AGE	3.149005
8	DIS	4.355469	8	DIS	4.333734
9	RAD	8.345247	9	RAD	2.942862
10	TAX	10.191941	10	PTRATIO	1.909750
11	PTRATIO	1.943409	11	LSTAT	2.860251
12	LSTAT	2.861881			

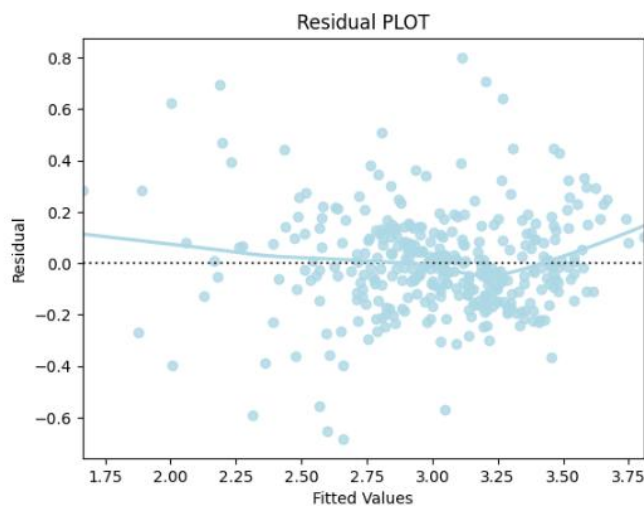
The left image shows the VIF of for each of the variables. As shown, the RAD and TAX values are above 5. This proves that there is multicollinearity between the two variables. To resolve this, the TAX variable (which had the highest VIF), was removed and gives the right image. As shown, most of the VIF values for the variables has decreased, especially the RAD which decreased to 2.94. In conclusion, by removing the TAX variable, multicollinearity has been removed.

Before creating the linear regression model, all the assumption has been checked.

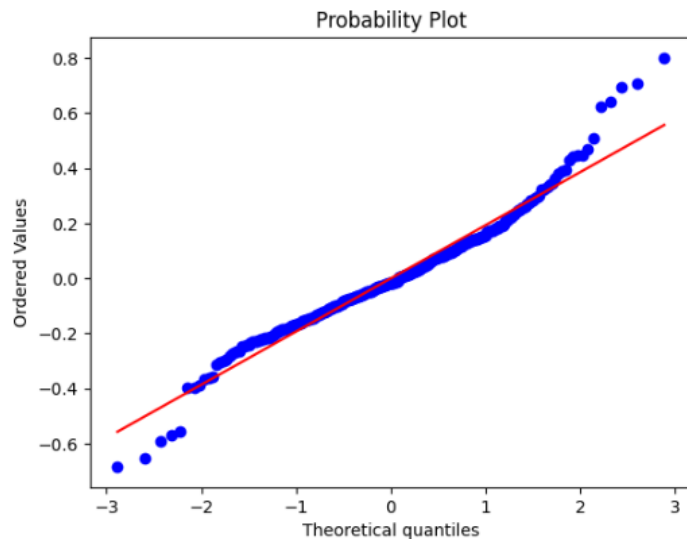
First, the mean value for residuals was checked and came out to $-2.53e-15$, which is very close to 0.

Second, when checking for homoscedasticity, the p-value came out as 0.3. This value is greater than the significance level of 0.05 which fails to reject null hypothesis of the data being homoscedastic.

Third, the residual plot was created to check if there ae no strong patterns for the residuals. As shown, the points here are all randomly scattered around zero.



Fourth, the probability plot was created to check for normal distribution. As shown, there is a strong normal distribution.



After clearing all assumptions, the model was created and the performance was tested. The results are the following.

	Data	RMSE	MAE	MAPE
0	Train	0.195504	0.143686	4.981813
1	Test	0.198045	0.151284	5.257965

Based on these numbers, while the RMSE is very similar, the MAE and MAPE values slightly increased on the test set from the training set. This implies that the model performs slightly worse on unknown data. The model score came out to a value of 0.769 and this implies that 76.9% of the variation in MEDV_log can be explained by the independent variables which is reasonably good.

Lastly, a cross-validation was applied to improve the model and get a more reliable model score. After 10 folds on the training data, the new model score came out to be 0.729. While this is slightly less than before, this drop isn't large enough to conclude that something went wrong, but instead, gives a more realistic estimate on how the model will perform for future unknown data.

Through this entire process of building a model, we were able to pinpoint what variables affect the price of houses by how much and find out that this model can be used to reliably predict unknown house prices. This model provides a very useable tool for businesses to accurately validate other house prices through trending data.

Conclusion and Recommendations

To summarize up all of the methods and findings

- Univariate analysis showed that each variable from the dataset had its own distribution and skewness
- The dependent variable MEDV was log transformed to better suit linear regression
- Bivariate analysis showed the correlation between the variables
- Heatmap from bivariate analysis showed that the variables RM and LSTAT are the most statistically significant in determining the house price (MEDV)
- VIF showed any multicollinearity and the biggest one (RAD-TAX) was removed to better suit the model
- Model was created with a score of 0.769
- Model was improved using cross-validation with 10 folds
- Final model was created with a score of 0.729

For businesses, some recommendations are the following.

- The number of rooms (RM) and the proportion of lower status people in the area (LSTAT) are the strongest determinants of the house price. More rooms will increase the price while more lower status people in the area will decrease the price. First, take into consideration these two feature when deciding the price.
- Be sure to check the other important attributes as well and know how they correlate. For example, higher crime rate will negatively affect the price.
- Improvement on certain variables can affect the higher the price. Having a police station near the house to decrease crime rate, planting more trees and greens which filters to air to reduce NOX concentration, or constructing more roads to shorten the distance to the employment stations, can all affect the house prices.
- The most common improvement used worldwide is renovations of houses to increase the number of rooms
- Don't focus on only the high-price houses, but on the lower-priced ones as well for buyers on a budget. While the price can be lowered due to certain variables, try to manage a variety of them. For example, some people might not care too much about the distance to employment centers, but care more about the NOX concentration that might affect their health
- Although some variables may negatively affect the price, the factor of 'personal preference' always come into play when making real-life decisions, so be ready for anything